EVALUATION

I. Choose the best answer

1. Gases deviate from ideal behavior at high pressure. Which of the following statement(s) is correct for non-ideality?

   a) at high pressure the collision between the gas molecule become enormous

   b) at high pressure the gas molecules move only in one direction

   c) at high pressure, the volume of gas become insignificant

   d) at high pressure the intermolecular interactions become significant

2. Rate of diffusion of a gas is

   a) directly proportional to its density

   b) directly proportional to its molecular weight

   c) directly proportional to its square root of its molecular weight

   d) inversely proportional to the square root of its molecular weight

3. Which of the following is the correct expression for the equation of state of van der Waals gas?

   (a) $\left(P + \frac{a}{n^2V^2}\right)(V - nb) = nRT$ (b) $\left(P + \frac{na}{n^2V^2}\right)(V - nb) = nRT$

   (c) $\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$ (d) $\left(P + \frac{n^2a^2}{V^2}\right)(V - nb) = nRT$

4. When an ideal gas undergoes unrestrained expansion, no cooling occurs because the molecules

   a) are above inversion temperature

   b) exert no attractive forces on each other

   c) do work equal to the loss in kinetic energy

   d) collide without loss of energy

5. Equal weights of methane and oxygen are mixed in an empty container at 298 K. The fraction of total pressure exerted by oxygen is

   (a) $1/3$ (b) $1/2$ (c) $2/3$ (d) $1/3 \times 273 \times 298$

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